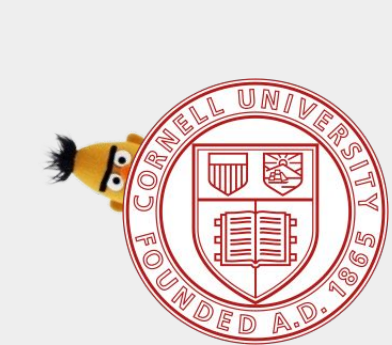




DistilBERT Re-Implementation

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Introduction

- LLMs like BERT (~110M parameters) exhibit strong NLP performance but are expensive to deploy (see figure 1 for a comparison of model sizes)
- Knowledge distillation: training a smaller “student” model to mimic a larger “teacher”
- DistilBERT is a distilled version of BERT - applies distillation during pre-training

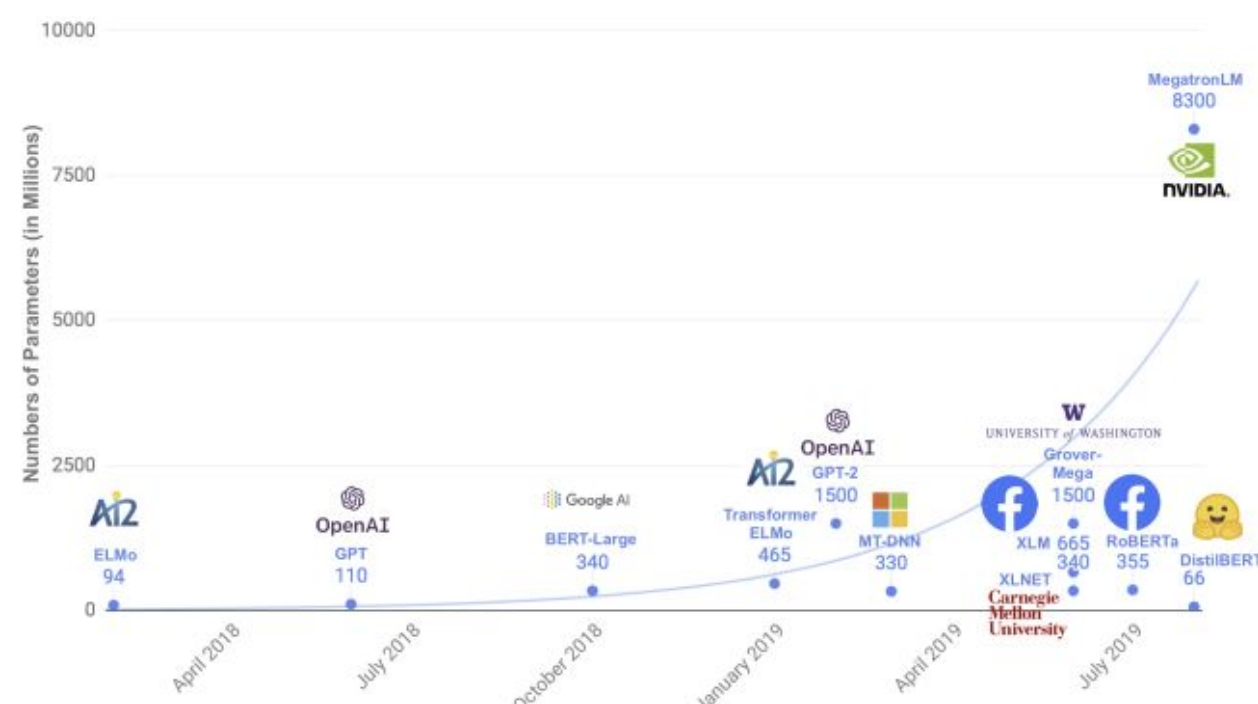


Figure 1. Pretrained language model sizes. Adapted from Sanh et al. (2019).

Background

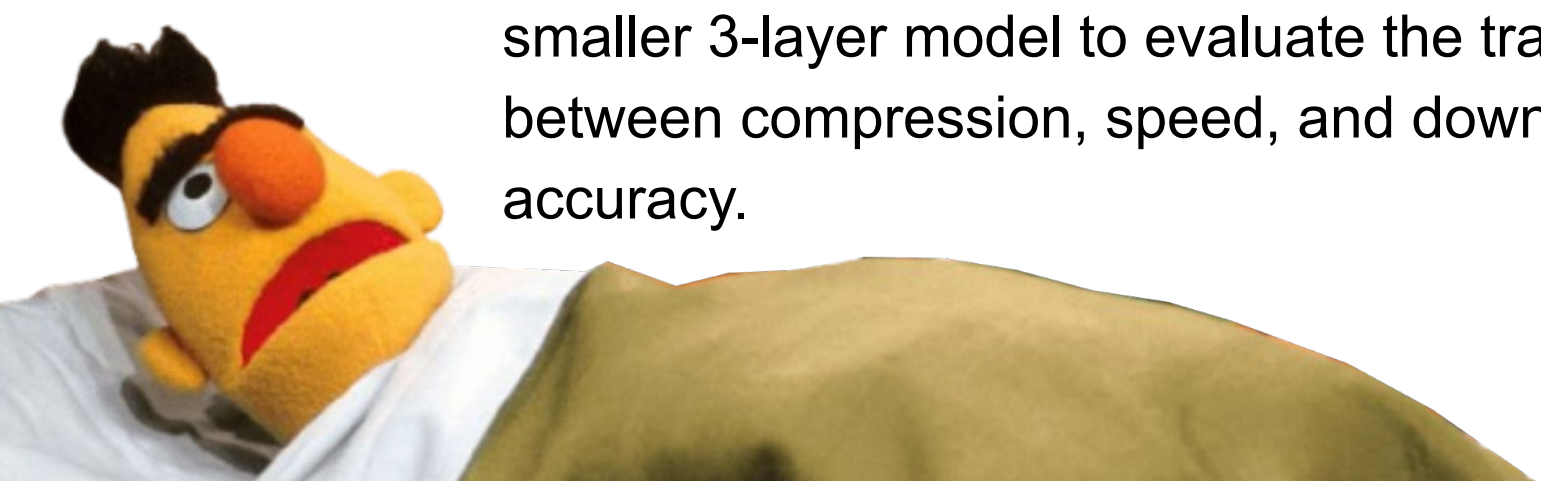
- BERT-base has about 109M parameters and 12 transformer layers.
- DistilBERT compresses BERT into a 6-layer student initialized from alternating BERT layers.
- Knowledge distillation transfers teacher behavior through soft-label predictions.
- Original paper claim: about 40% fewer parameters, about 60% faster inference, while retaining most BERT performance.
- Our goal was to test whether these trends hold in a smaller compute setting.

Research Goal

We aimed to reproduce four key results from the DistilBERT paper:

- Parameter reduction: compare BERT-base, 6-layer student, and 3-layer student.
- Downstream accuracy: fine-tune on IMDb sentiment classification.
- Inference speedup: measure teacher vs. student forward-pass latency.
- Ablation: remove the soft cross-entropy distillation loss and compare performance.

- We also extended the experiment: trained a smaller 3-layer model to evaluate the tradeoff between compression, speed, and downstream accuracy.



Methodology

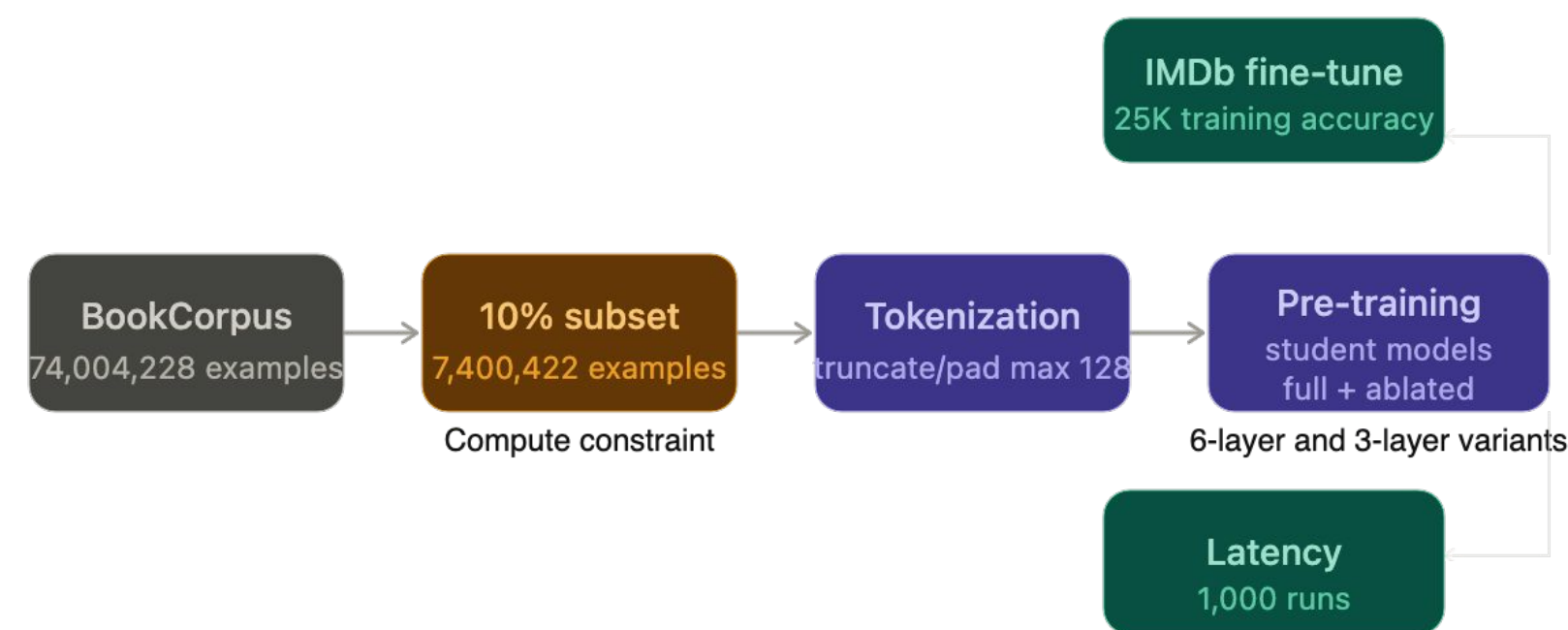


Figure 2: Training and Evaluation Pipeline

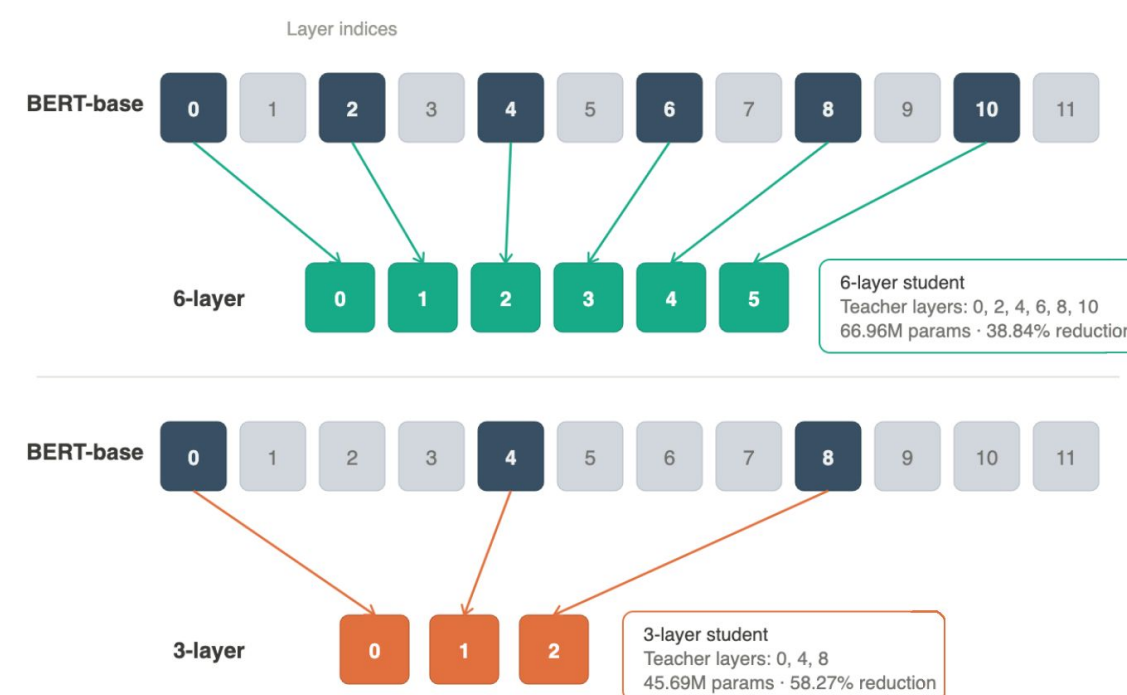


Figure 3: Bert-to-Student Layer Initialization

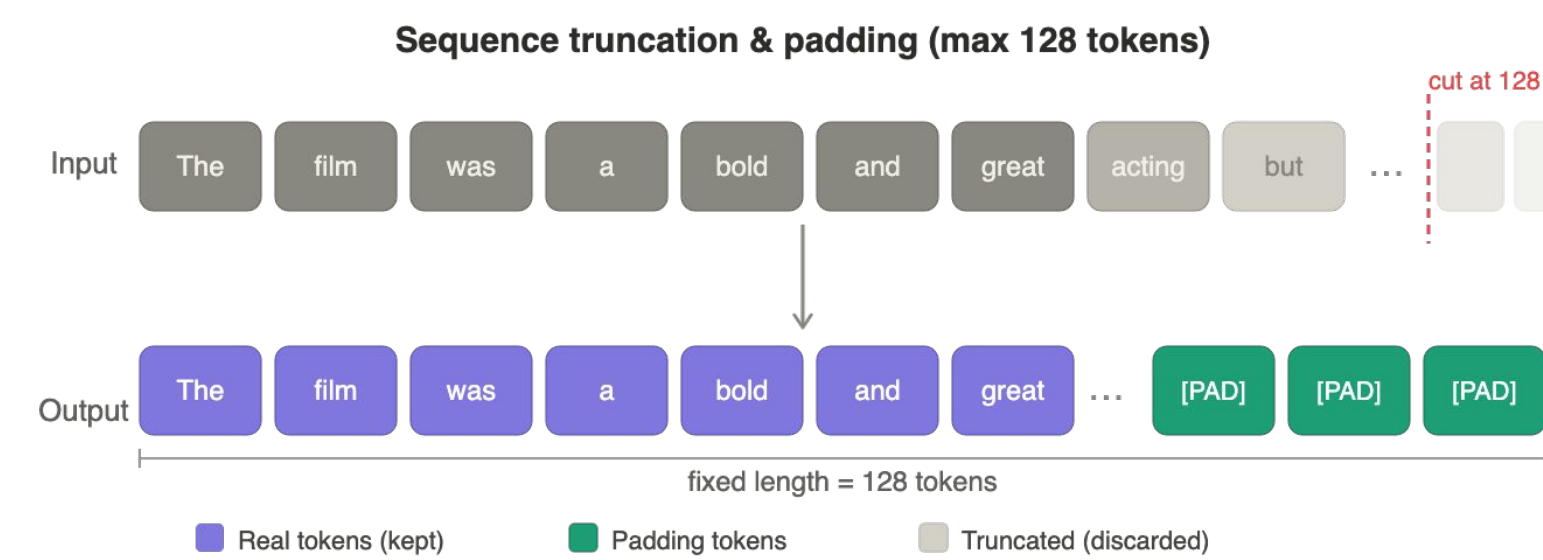


Figure 4: Sequence Truncation and Padding

Model Architecture & Setup

- Figure 2 shows the training and evaluation pipeline, including BookCorpus pretraining, IMDb fine-tuning, and inference-latency testing.
- As shown in Figure 3, the 6-layer and 3-layer students were initialized by copying certain layers from the BERT model
- After initialization, both student models were pretrained with masked language modeling, KL soft-label distillation, and cosine teacher-student alignment losses.

Data And Compute/Limitations:

- Pre-training used 10% of BookCorpus due to compute constraints.
- Sequences were truncated/padded to max length 128 as shown in Figure 4.
- IMDb fine-tuning used the 25K training split for sentiment classification.
- Inference latency was averaged over 1,000 forward passes.

Results

Model	Parameters	Reduction	IMDb accuracy	Latency
BERT-base teacher	109.48M	0.00%	93.46% (paper)	7.55 / 7.34 ms
6-layer student, full loss	66.96M	38.84%	91.42%	4.24 ms
6-layer student, CE ablated	66.96M	38.84%	91.40%	same architecture; not separately benchmarked
3-layer student, full loss	45.69M	58.27%	90.01%	2.38 ms
3-layer student, CE ablated	45.69M	58.27%	90.12%	same architecture; not separately benchmarked

Table 1: Reimplementation Results Summary

Inference Speed:

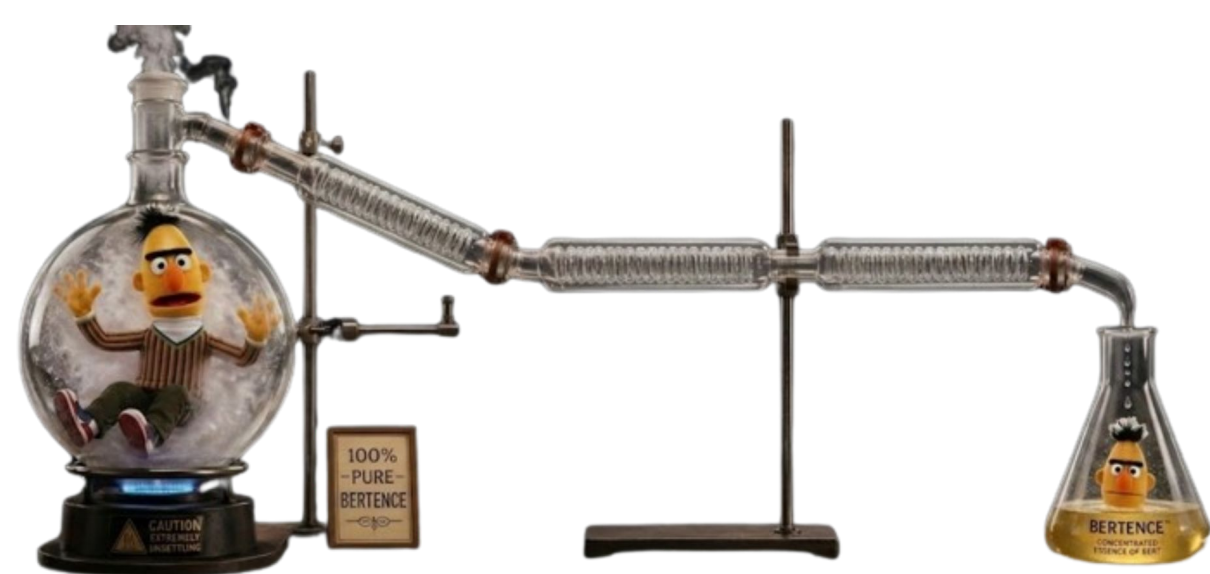
- As demonstrated in Table 1, the 6-layer student model had an average inference speed of 4.24 ms compared to the teacher model's 7.55 ms, a 1.78x speedup.
- In addition, the 3-layer student model achieved an average inference speed of: 2.38 ms compared to the teacher model's 7.34 ms, a 3.08x speedup.

Ablation:

- As visualized in Figure 5, full-loss models generally converge faster and smoother than the CE-ablated models, supporting distillation term helps student learn more efficiently.

Conclusion

- Successfully reproduced the core DistilBERT findings, confirming that student models significantly reduce size and latency while maintaining strong downstream performance.
- While absolute IMDb accuracy was lower than the original paper (likely due to pre-training data constraints), the accuracy-to-size ratio remains high.



- The 3-Layer Model offered the most significant speedup and parameter reduction while maintaining accuracy levels surprisingly close to the 6-layer model
- For simple downstream tasks like sentiment classification, aggressive compression can be used without a substantial loss in representation quality.
- Results demonstrate that Knowledge Distillation is a highly effective strategy for deploying NLP models in environments with limited compute resources.

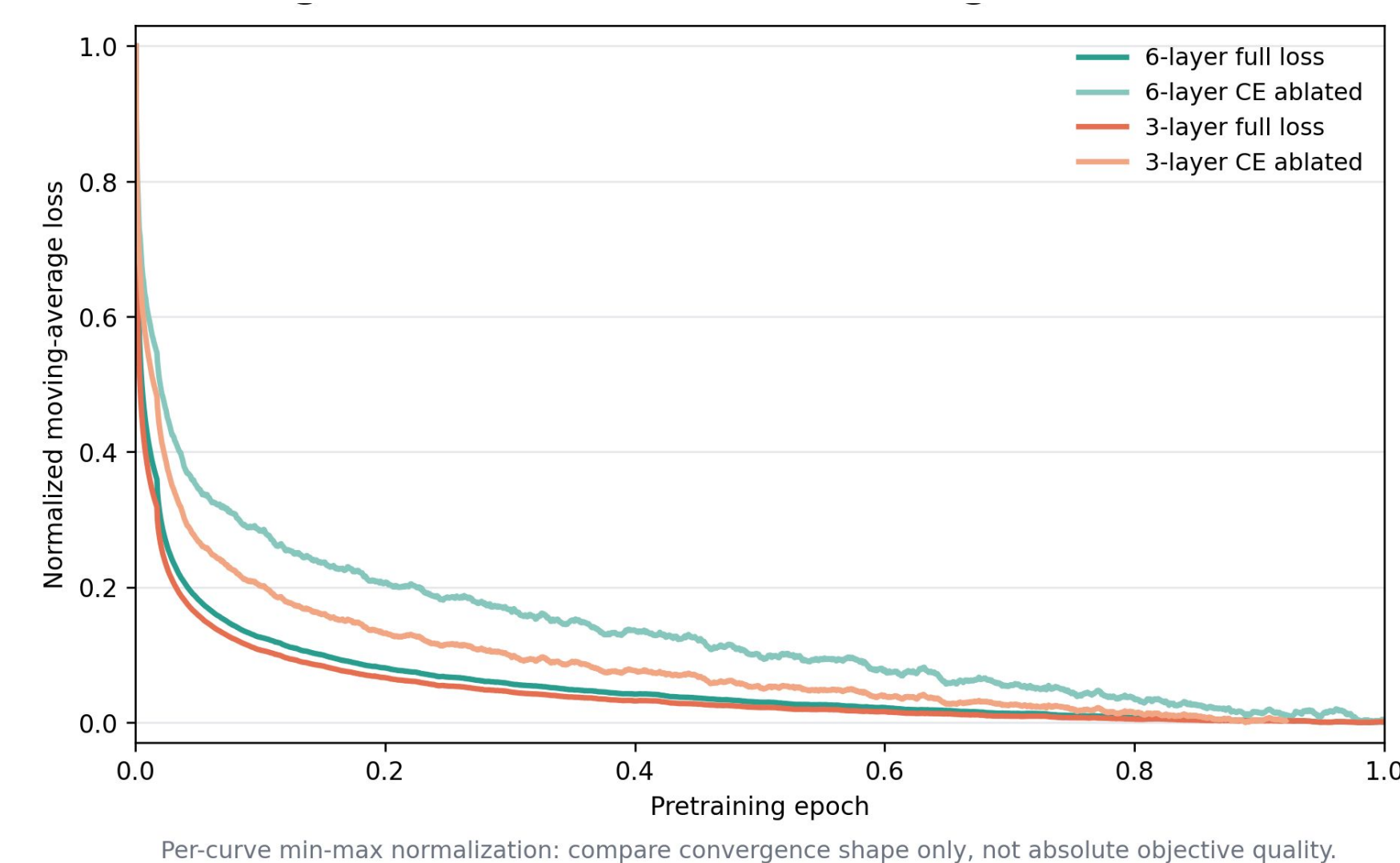


Figure 5: Normalized Pretraining Loss Trends

Future Work

- Train on larger pre-training data, including Wikipedia.
- Run multiple random seeds.
- Evaluate on GLUE tasks.
- Separate ablations for MLM loss, and cosine loss.
- Explore even smaller students or layer-selection strategies.

References

- Sanh, Victor, et al. DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter. 2019. arXiv, arXiv:1910.01108.